## Models

Side A

Base (adapter off)

Side B

apertus-nash / step-000517

Newest checkpoint is first in the list.

## Sampling

Temperature (0 = greedy)

0.60

top_p

0.95

Max new tokens

2816

Seed (-1 = random)

-1

## Prompt

Thinking mode

System prompt (same for both sides)

You are Apertus Nash, a general-purpose AI assistant built on the open Apertus model, with particular strengths in mathematics, statistics

## GPU

101 / 131 GB in use

Loaded: side A, side B

Unload both models

# Apertus demo: base vs. checkpoint

A = Base (adapter off) · B = apertus-nash / step-000517 · cuda · both sides generate at the same time with the same settings

Side by side | Benchmark

Send each message to

New chats

Both | Side A only | Side B only

## A: Base

Base (adapter off)

hey

thinking (134 tok)

Hello! How can I assist you today?

new 146 tok · 14s · 10.1 tok/s · stop token <|assistant_end|>

Can you teach me algebra? The very basics step by step, start with step one

thinking (522 tok)

**Step 1 – What is Algebra?**
Algebra is the branch of mathematics that uses symbols (usually letters) to represent numbers and quantities. Instead of only working with concrete numbers, we use variables to describe unknown values and write rules for how those values interact.

**Step 2 – What are Variables?**
A variable is a symbol (like $x, y, z$) that stands for a number we don't yet know. For example, if we let $x$ represent "the number of apples," we can write an expression such as $2x$ meaning "twice the number of apples."

**Step 3 – What is an Expression?**
An algebraic expression is a combination of numbers, variables, and operations (addition, subtraction, multiplication, division).
Examples:
$-3+5$ (simple numeric expression)
$-2x-7$ (expression with a variable)
$-4(y+2)$ (expression with parentheses and multiplication)

**Step 4 – What is an Equation?**
An equation is a statement that two expressions are equal. It always contains an equals sign (=).
Example: $2x-3=7$ tells us that the left-hand side equals the right-hand side.

## B: step-000517

apertus-nash / step-000517

hey

thinking (6 tok)

Hi! How can I help you today?

new 19 tok · 5s · 4.1 tok/s · stop token <|assistant_end|>

Can you teach me algebra? The very basics step by step, start with step one

thinking (22 tok)

Sure! Let's start with the very basics and go step by step.

**Step 1: What are numbers?** Numbers are the things we count or measure. For example, 2 apples, 5 meters, or -3 degrees.

**Step 2: What are variables?** A variable is a symbol that stands for a number. Common symbols are x, y, and z. Think of it like a placeholder that can be any number.

**Step 3: What is an expression?** An expression is a group of numbers and symbols combined. For example:

- 3 + 5
- 2x - 7
- 4y + 10

**Step 4: What is an equation?** An equation says two expressions are equal. It has an equals sign (=). For example:

- 2 + 3 = 5
- 2x + 5 = 11

**Step 5: Solving an equation** To solve an equation means to find the value of the variable that makes the equation true. Let's solve 2x + 5 = 11.

**Step 6: Subtract 5 from both sides** 2x + 5 - 5 = 11 - 5 This simplifies to: 2x = 6

**Step 5 – What does it mean to "solve an equation"?**
To solve an equation means to find the value(s) of the variable that make the equation true.

**Step 6 – Solving a Simple Linear Equation (Example)**
Take the equation: $2x - 3 = 7$

1. **Isolate the term with the variable** – Add 3 to both sides:
   $2x - 3 + 3 = 7 + 3 \rightarrow 2x = 10$
2. **Solve for x** – Divide both sides by 2:
   $2x \div 2 = 10 \div 2 \rightarrow x = 5$

Check: Plug $x = 5$ back into the original equation:
$2(5) - 3 = 10 - 3 = 7$ ✓

**Step 7 – Basic Properties of Algebra**
These rules help us manipulate equations without changing their truth:

| Property | Description |
|---|---|
| **Commutative** | $a + b = b + a$ and $ab = ba$ |
| **Associative** | $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ |
| **Distributive** | $a(b + c) = ab + ac$ |

**Step 8 – Practice**
Try solving these simple equations on your own, then compare your steps to the example above.

1. $x + 4 = 9$
2. $3y - 2 = 7$
3. $2z \div 4 = 3$

When you're ready, feel free to ask for hints or a full solution for any of these!

new 1275 tok · 141s · 9.1 tok/s · stop token <|assistant_end|>

make me a consice table of the things I should learn in linear algebra

› thinking (496 tok)

| Topic | Key Concepts / What to Learn |
|---|---|
| Vectors & Vector Operations | Addition, scalar multiplication, dot product, magnitude, unit vectors |
| Linear Independence & Basis | Determining independence, spanning sets, constructing a basis for a subspace |
| Dimension & Rank | Computing dimension of subspaces, rank of a matrix (row/column rank) |

**Step 7: Divide both sides by 2** $2x / 2 = 6 / 2$
This simplifies to: $x = 3$

**Step 8: Check your answer** Plug $x = 3$ back into the original equation: $2(3) + 5 = 6 + 5 = 11$ It works! So $x = 3$ is the correct solution.

That's the very basics of algebra. Would you like to practice more or move on to something else?

new 408 tok · 81s · 5.0 tok/s · stop token <|assistant_end|>

make me a consice table of the things I should learn in linear algebra

› thinking (27 tok)

Sure! Here's a concise table of key topics in linear algebra:

| Category | Key Topics |
|---|---|
| **Vectors** | • Vector addition and scalar multiplication<br>• Magnitude and direction<br>• Unit vectors<br>• Linear combinations |
| **Matrices** | • Matrix addition and multiplication<br>• Identity and zero matrices<br>• Transpose<br>• Inverses |
| **Systems of Equations** | • Gaussian elimination<br>• Row-echelon form<br>• Consistent/inconsistent systems |
| **Linear Independence** | • Span and basis<br>• Determining dependence |
| **Eigenvalues & Eigenvectors** | • Definition and calculation<br>• Diagonalization |
| **Orthogonality** | • Dot product and orthogonality<br>• Orthogonal projections |
| **Transformations** | • Linear transformations<br>• Change of basis |
| **Applications** | • Least squares<br>• Markov chains<br>• Data analysis |

Would you like more details on any of these topics?

new 256 tok · 52s · 5.0 tok/s · stop token <|assistant_end|>

| Topic | Key Concepts / What to Learn |
|---|---|
| Matrices & Matrix Operations | Matrix addition, multiplication, transpose, inverse, determinant |
| Solving Linear Systems | Gaussian elimination, row-reduction, interpreting solution sets (unique, infinite, none) |
| Determinants | Definition, properties, calculating determinants, Cramer's rule for solving systems |
| Eigenvalues & Eigenvectors | Characteristic polynomial, finding eigenvalues/eigenvectors, diagonalization |
| Inner Products & Orthogonality | Dot product, norms, orthogonal sets, orthogonal projections, Gram-Schmidt process |
| Linear Transformations | Matrix representation, kernel (null space), image (column space), change-of-basis |
| Advanced Topics | Diagonalization, Jordan canonical form, Singular Value Decomposition (SVD), applications in data science, computer graphics, physics, engineering |

Feel free to expand any row as you progress and practice computational

Message the models…

new 773 tok · 87s · 8.8 tok/s · stop token <|assistant_end|>